

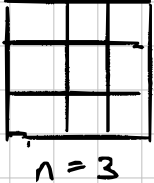
Assignment 4

Problem 3.6

Given: $n \times n$ warehouse grid

Part A: Propositional Encoding Cost

- 1) If $n = 3$ as an example, the adjacency sentences will be:



$(1,1), (1,2), (2,1)$

$(2,1), (2,2), (3,1)$

$(3,1), (2,1), (3,2)$

$(1,2), (1,1), (2,2), (1,3)$

$(2,2), (2,1), (1,2), (2,3), (3,2)$

$(3,2), (3,1), (2,2), (3,3)$

$(1,3), (1,2), (2,3)$

$(2,3), (1,3), (2,2), (3,3)$

$(3,3), (3,2), (2,3)$

There are 9 adjacency sentences

$\Rightarrow n^2$ number of adjacency sentences,

$\therefore$ as n grows, the number of adjacency sentences grows exponentially.

At $n = 100$, number of adjacency sentences are $100^2 = 10000$

2) $D_{2,1} \Rightarrow C_{1,1}, C_{2,2}, C_{3,1}$

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$D_{1,2} \Rightarrow C_{1,1}, C_{2,2}, C_{1,3}$

$D_{2,2} \Rightarrow C_{2,1}, C_{1,2}, C_{2,3}, C_{3,2}$

$D_{3,2} \Rightarrow C_{3,1}, C_{2,2}, C_{3,3}$

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$D_{2,3} \Rightarrow C_{1,3}, C_{2,2}, C_{3,3}$

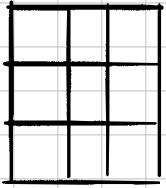
$D_{3,3} \Rightarrow C_{3,2}, C_{2,3}$

There are 8 damage sentences because $(1,1)$ is safe.

$\Rightarrow n^2 - 1$ number of damage sentences

At $n = 100$, number of damage sentences based on the creating rule $= 100^2 - 1 = 9999$

3) Safety Rule.



Since $(1,1)$ is always safe,
then number of safety sentences
needed using the safety rule is

$$n^2 - 1$$

Part B : FOL Comparison

- 1) Adjacency : Each square has at least one adjacent

For every square L , there exists at
least one adjacent square L' :

$$\forall L \exists L' \text{Adjacent}(L, L')$$

Only one sentence needed regardless of n .

- 2) $\forall L \text{Creaking}(L) \Leftrightarrow \exists L' \text{Adjacent}(L, L') \wedge \text{Damaged}(L')$

For every location, L , that creaks, will only
happen iff
there exists an adjacent location to L , L' ,
and that L' is damaged.

- 3) $\forall L \text{Safe}(L) \Leftrightarrow \neg \text{Damaged}(L) \wedge \neg \text{Forklift}(L)$
or
 $\forall L \text{Safe}(L) \Leftrightarrow \neg (\text{Damaged}(L) \vee \text{Forklift}(L))$

Part C : Translation Challenge

"There is exactly one damaged floor section in the entire warehouse"

$$\exists! L \text{ Damaged}(L) \wedge \forall L' \text{ Damaged}(L') \Rightarrow L' = L$$

There exists only one damaged floor and for every detected damage floor, L' , then that must be L .

Part D : Practical Implications

1. For a 100×100 warehouse with 10,000 locations, a propositional encoding practice would require 100^2 sentences.

Does not matter that there are 10,000 locations if all locations are 100×100 and can then use the same set of propositional encoding.

However, it is still not elegantly scalable as n increases.

2. FOL cannot be used in second order properties. So the above scenario, where the 10,000 locations will be an example.